Executive Briefing: FlashLight-BPM Setup Context and Mobile PPG Application Architecture

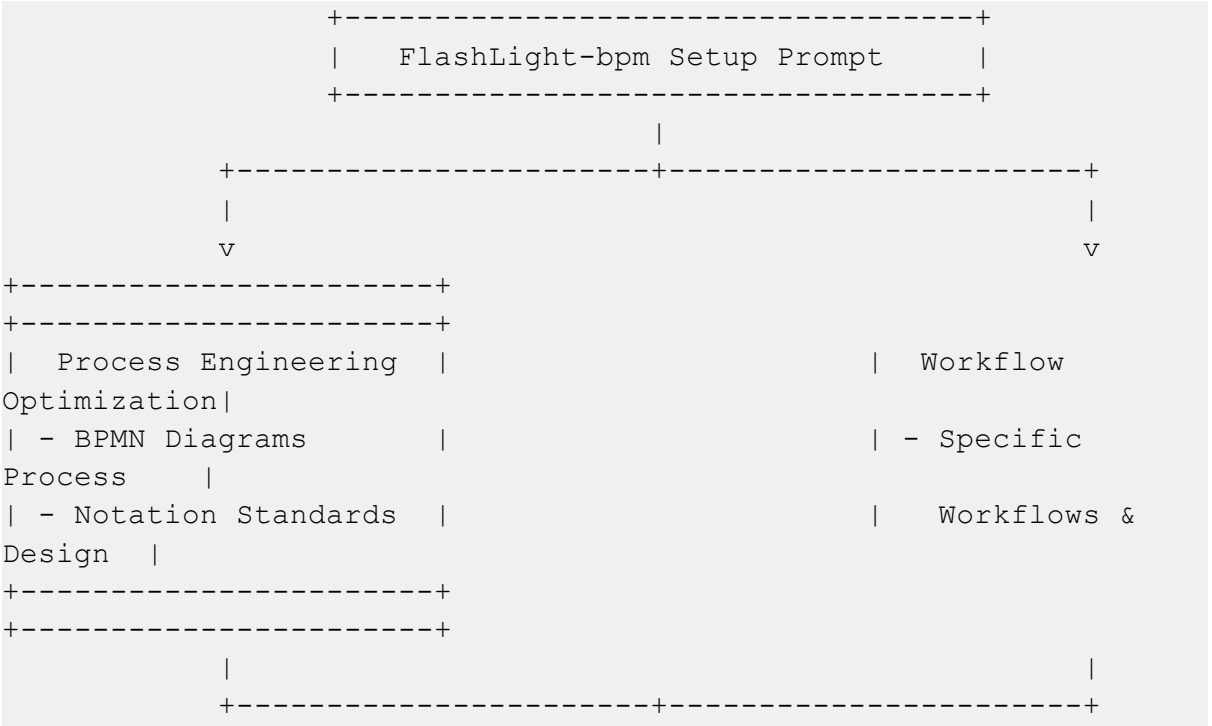
Executive Summary

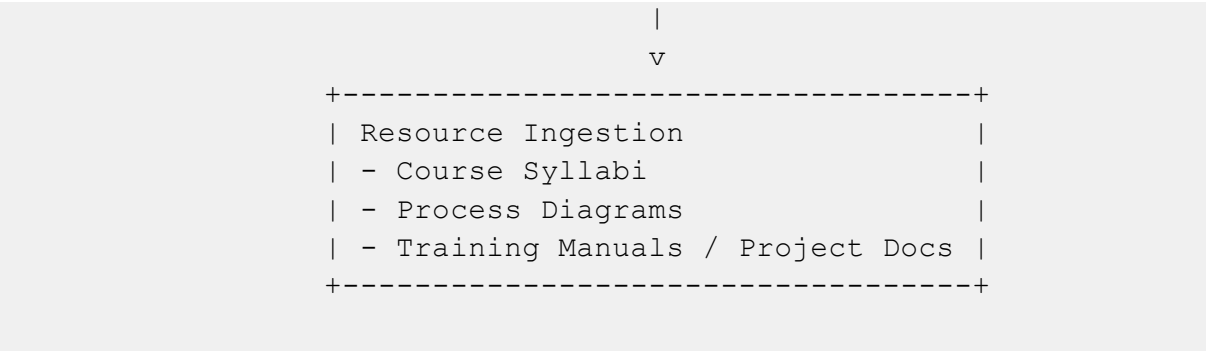
This briefing document synthesizes technical specifications and setup workflows derived from the **FlashLight-bpm** ecosystem. The sources encompass conversational AI setup assistants for process management learning alongside the source code evolution of flashligh-bpm-app—a cross-platform Flutter application (Android/iOS) designed to estimate heart rate in real time via smartphone camera photoplethysmography (PPG).
The codebase reflects a progressive engineering evolution across four major repository milestones (master, v1, v2, and v3). The core system captures raw optical signals using the rear camera lens and flashlight torch, processes green-channel intensity through a multi-stage time-invariant Digital Signal Processing (DSP) pipeline, evaluates tissue optical density via a robust finger-presence state machine, and outputs filtered pulse wave diagnostics alongside CSV session data.
Key technical milestones include:

- Transitioning from variable time-delta filtering to uniform **30 Hz linear interpolation** resampling (v1).
- Introducing **stream interruption recovery** for frame drops exceeding 100 ms (v2).
- Expanding optical skin-tone classification boundaries to accommodate broader hardware variations (v2).
- Adding **interactive UX guidance** , automatic session timeouts, and locked result summaries upon scan completion (v3).

1. FlashLight-BPM Notebook Setup & Goal Customization

Conversational setup assistants for the **FlashLight-bpm** environment focus on onboarding users through primary goal selection and training asset ingestion.



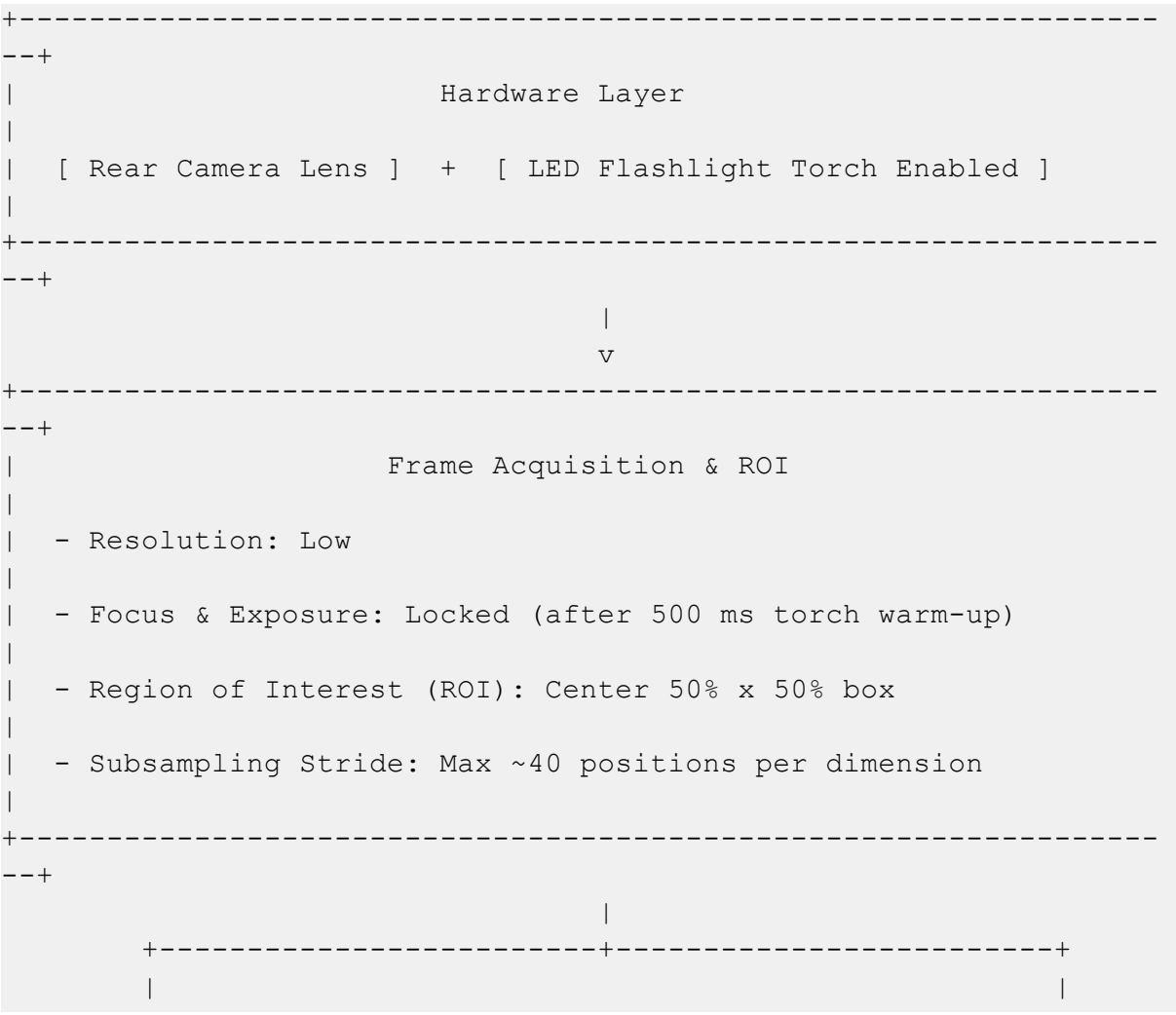


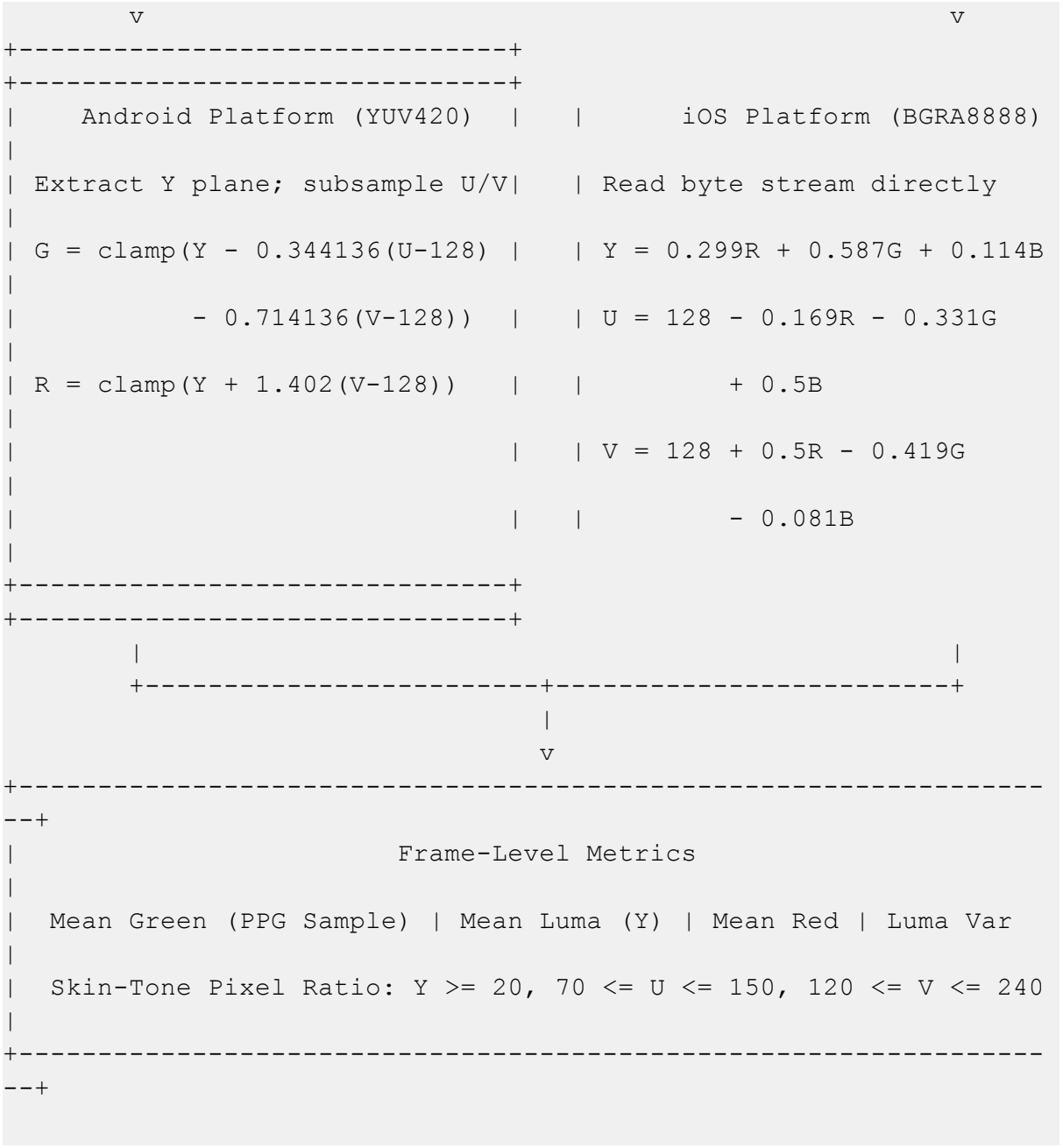
Primary Setup Axes

- **BPMN Mastery vs. Process Optimization:** Users select between mastering formal Business Process Model and Notation (BPMN) diagrams/standards or designing and optimizing domain-specific business processes.
- **Context Ingestion:** The setup routine ingests uploaded course syllabi, process models, project documentation, or corporate training manuals to tailor interactive prompts.

2. Technical Architecture: Mobile PPG Signal Acquisition

The flashligh-bpm-app relies on non-invasive optical PPG to measure pulsatile changes in microvascular tissue. It uses Flutter for cross-platform support across Android and iOS.





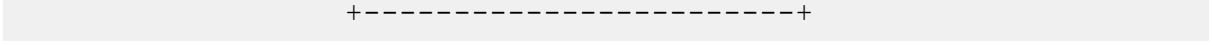
Frame Metrics Computation

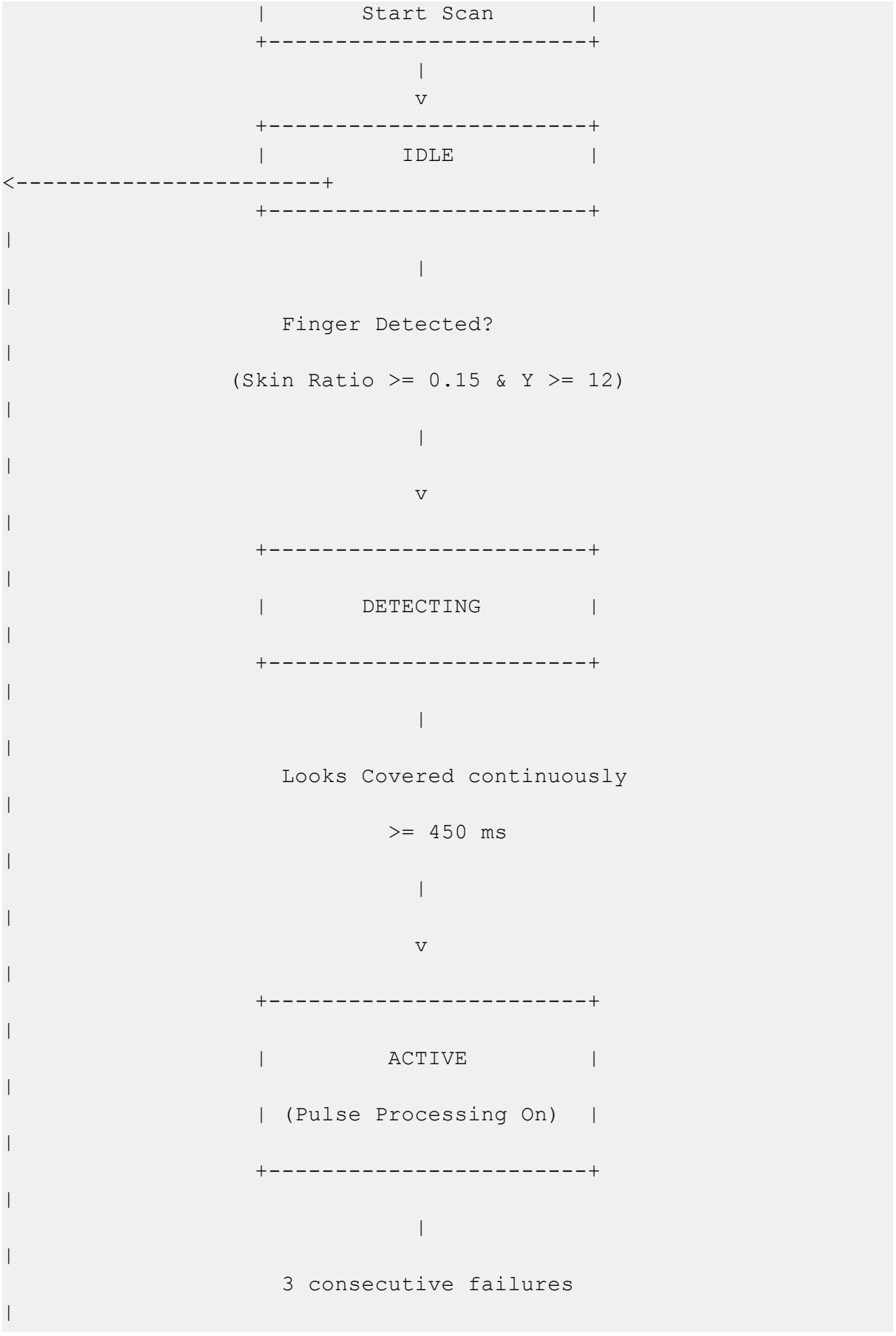
Across all sampled ROI pixels, the app computes:

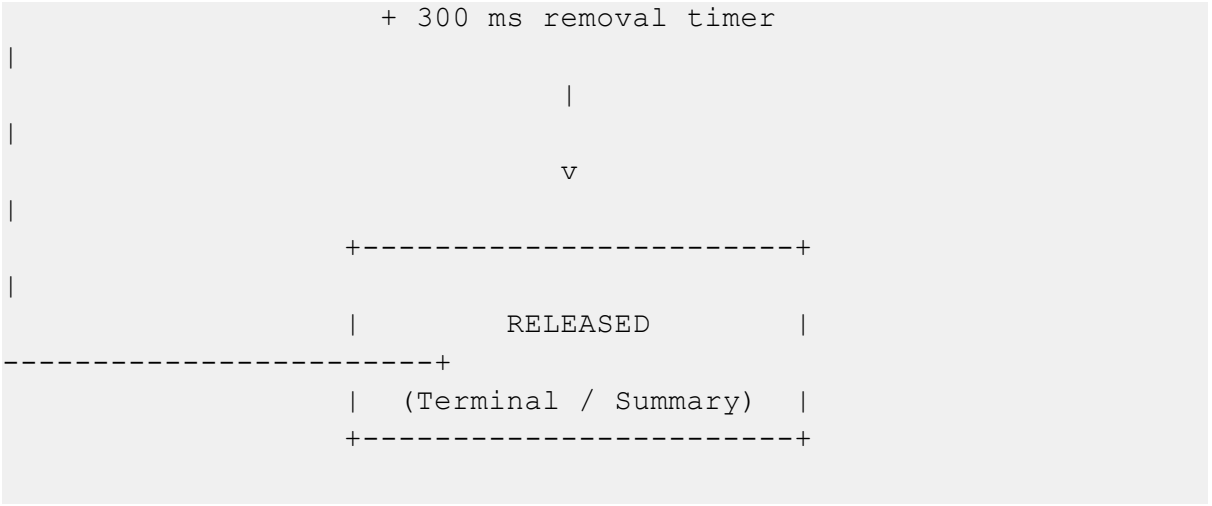
1. **Mean Green ($\text{mean } G$):** Primary raw PPG amplitude signal.
2. **Mean Luma ($\text{mean } Y$):** Average frame brightness.
3. **Luma Variance:** $\text{mean}(Y^2) - (\text{mean } Y)^2$, clamped to a lower bound of zero.
4. **Skin-Tone Ratio:** Fraction of sampled pixels where $Y \ge 20$, $70 \le U \le 150$, and $120 \le V \le 240$ (expanded from $V \le 195$ in earlier builds).

3. Finger-Presence Optical State Machine

To isolate true pulse data from movement artifacts or ambient air scans, an internal optical gate enforces strict state transitions.




Start Scan
IDLE
Finger Detected?
(Skin Ratio >= 0.15 & Y >= 12)
DETECTING
Looks Covered continuously
>= 450 ms
ACTIVE
(Pulse Processing On)
3 consecutive failures

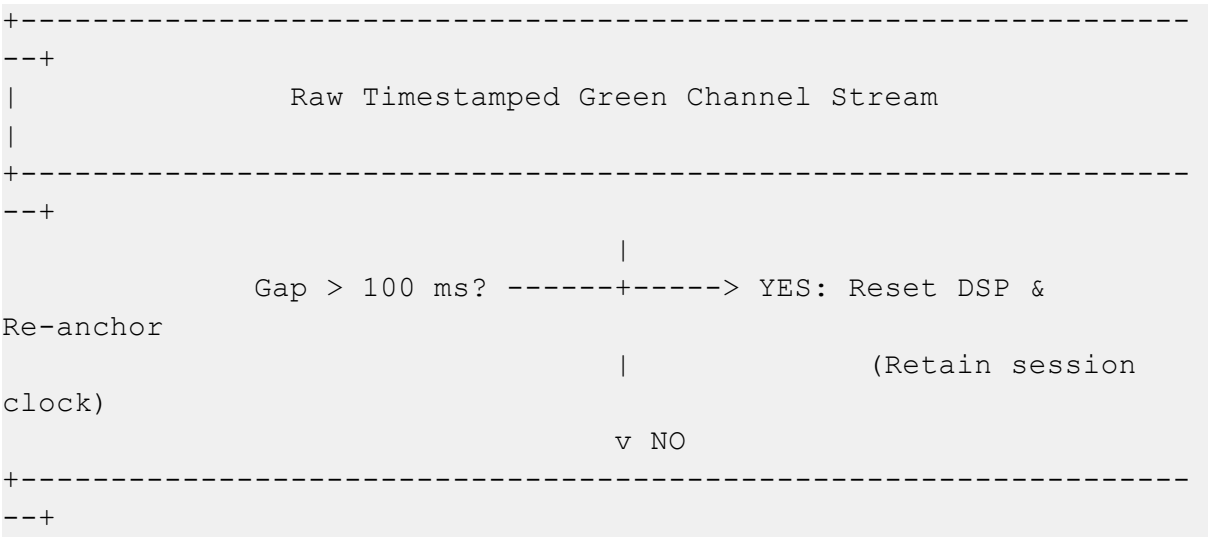


State Machine Rules and Baseline Calibration

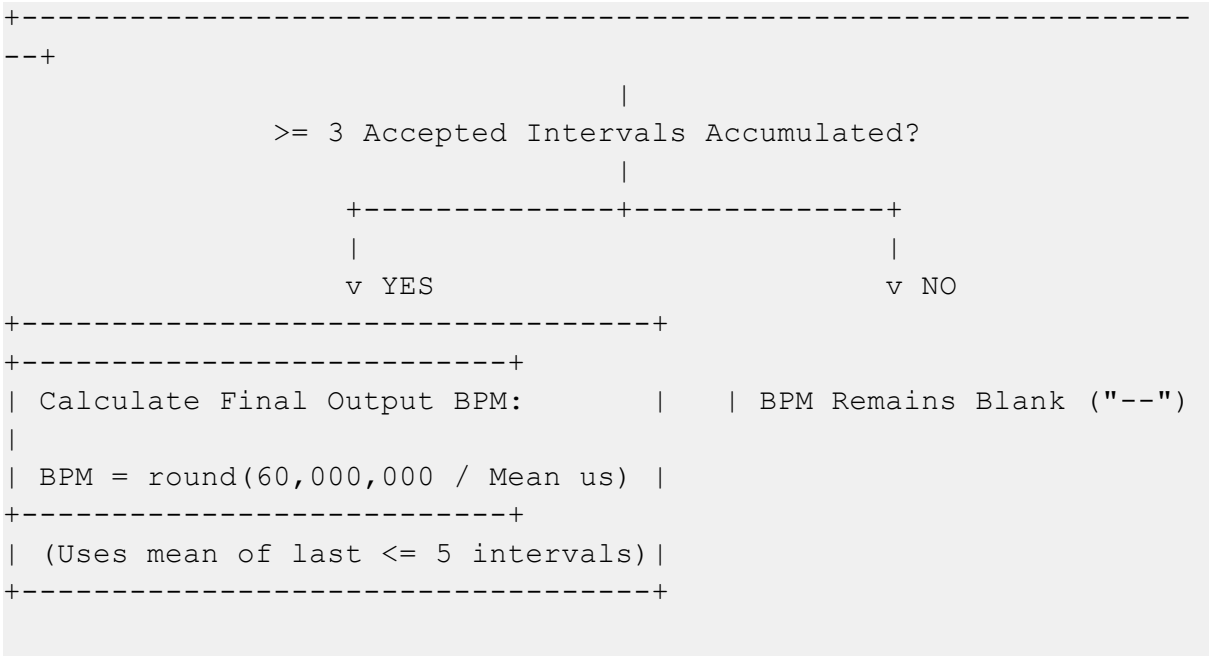
- **Ambient Calibration:** Upon launch, the first second (requiring at least 8 frames) measures background lighting with the lens uncovered to set baseline luma ($\text{mean } Y_{\text{base}}$) and luma variance ($\text{var } Y_{\text{base}}$).
- **Finger Detected Criterion:** Skin-tone ratio ≥ 0.15 AND mean luma ≥ 12 .
- **Looks Covered Criterion:** Requires all three conditions:
- Skin-tone ratio ≥ 0.55
- Mean luma ≥ 12
- **Either** absolute luma shift $|Y - Y_{\text{base}}| \ge \max(5, 8\% \times Y_{\text{base}})$ **OR** current luma variance $\le 75\% \times \text{var } Y_{\text{base}}$.
- **Debouncing and State Switches:**
- **To ACTIVE:** The finger must satisfy "Looks Covered" continuously for $\ge 450\text{ ms}$.
- **To RELEASED:** If coverage checks fail in ACTIVE state, the error counter debounces up to 3 consecutive frames before starting a 300 ms removal timer. Exceeding 300 ms moves the state to RELEASED.

4. Digital Signal Processing & Pulse Calculation Pipeline

Frames marked as ACTIVE enter the heart rate measurement processor.



```
|                  30 Hz Uniform Grid Resampler
|
|           Linear interpolation between camera timestamps
|
+-------------------------------------------------------------------
--+
                                    |
                                    v
+-------------------------------------------------------------------
--+
|               2-Stage Time-Invariant IIR Bandpass
|
|  - High-Pass Cutoff: 0.75 Hz (Alpha_HP = RC_HP / (RC_HP + 1/30))
|
|  - Low-Pass Cutoff:  3.50 Hz (Alpha_LP = (1/30) / (RC_LP +
1/30)) |
+-------------------------------------------------------------------
--+
                                    |
                                    v
+-------------------------------------------------------------------
--+
|                     Peak & Trough Extrema
|
|  - Local window: Preceding 500 ms (latest 3 samples minimum)
|
|  - AC/DC Prominence Check:
|
|    Prominence / DC >= max(0.1%, 18% * Local Range / DC)
|
+-------------------------------------------------------------------
--+
                                    |
                                    v
+-------------------------------------------------------------------
--+
|                 Pulse Interval Validation Logic
|
|  - Ignore interval if < 285 ms (~211 BPM upper limit)
|
|  - Clear buffer & BPM if interval > 1350 ms (~44 BPM lower
limit) |
|  - Interval 1: Accepted unconditionally
|
|  - Interval 2: Must be within 35% of Interval 1
|
|  - Interval 3+: Must be within 25% of retained median interval
|
```



Signal Quality Diagnostics

Signal quality is evaluated across a rolling 3-second (up to 90 resampled points) window by computing the AC peak-to-peak amplitude relative to the raw green DC mean ($\text{AC/DC} = (\max(\text{filtered}) - \min(\text{filtered})) / \text{mean raw green}$): $\text{Quality Classification} = \begin{cases} \text{ACQUIRING} & \text{if sample count} < 30 \\ \text{POOR} & \text{if } \text{AC/DC} < 0.2% \\ \text{GOOD} & \text{if } \text{AC/DC} \ge 1.0%, \ge 3 \text{ accepted intervals, and no active rejections} \\ \text{FAIR} & \text{if } \text{AC/DC} \ge 0.2% \text{ and conditions for GOOD are not met} \end{cases}$

5. Architectural Version Evolution (master to v3)

The application evolved across four main code revisions, addressing resampling stability, signal noise, platform variations, and user experience.

Subsystem / Feature	Base Commit (master)	Version 1 (v1)	Version 2 (v2)	Version 3 (v3)
Grid Resampling	None (Variable Δt per raw camera frame)	Fixed 30 Hz uniform grid via linear interpolation	Fixed 30 Hz uniform grid via linear interpolation	Fixed 30 Hz uniform grid via linear interpolation
Frame Drop Recovery	None	None	Resets DSP/BPM state on frame gaps $> 100\text{ ms}$	Resets DSP/BPM state on frame gaps $> 100\text{ ms}$
Filter Model	Variable Δt dynamic IIR bandpass	Time-invariant 0.75–3.5 Hz 2-stage IIR	Time-invariant 0.75–3.5 Hz 2-stage IIR	Time-invariant 0.75–3.5 Hz 2-stage IIR
Prominence Checking	Absolute filtered threshold (≥ 0.08)	Relative AC/DC ratio relative to local signal range	Relative AC/DC ratio relative to local signal range	Relative AC/DC ratio relative to local signal range
Skin Chroma Bounds	$120 \le V \le 195$	$120 \le V \le 195$	$120 \le V \le 240$	$120 \le V \le 240$
Directory Creation	Basic file write	Basic file write	Recursive directory creation (data/)	Recursive directory creation (data/)

Session UX & Feedback,Instant exit upon release; endless timer,Instant exit upon release; endless timer,Instant exit upon release; endless timer,"20s initial timeout, 5s poor signal visual hints, frozen Scan Complete screen"

Detailed Branch Key Changes

Base Commit (master)

- Initial Flutter implementation.
- Uses dynamic time deltas (Δt clamped between $1/120\text{ s}$ and $1/5\text{ s}$) for IIR filter poles.
- Employs static amplitude prominence thresholds (≥ 0.08) and fixed 90-sample window counts rather than time-anchored grids.

Version 1 (v1)

- Anchors processing on a fixed 30 Hz uniform interpolation grid.
- Stabilizes IIR bandpass coefficients regardless of camera frame rate variations.
- Replaces absolute signal thresholds with relative AC/DC peak prominence checks.
- Introduces a 3-frame debounce check before running the 300 ms removal timer.

Version 2 (v2)

- Introduces recovery from camera stream drops: raw camera frame delays exceeding 100 ms reset the resampler, IIR state, rolling waveform, and pulse interval buffers, re-anchoring the grid to the current frame.
- Expands the skin-tone threshold upper limit ($V \le 240$) to improve finger detection across diverse skin tones and device camera profiles.
- Ensures robust CSV logging by recursively creating the parent data/ directory before opening heart_rate_samples.csv.

Version 3 (v3)

- **Visual Target Overlay:** Adds a center alignment target to the live camera preview to guide placement over both the lens and the flashlight torch.
- **Scan Complete Summary Screen:** Replaces immediate screen dismissal on finger removal with a frozen summary screen showing final BPM, signal quality rating, session duration, and sample counts. Tapping "Finish" manually triggers this view.
- **Timeout Mechanics:** Enforces a 20-second connection timeout: if ACTIVE coverage is not achieved within 20 seconds of launching the measurement view, the camera and torch shut off and return the user to the home screen with a notification.
- **Dynamic Placement Guidance:** Displays visual guidance for finger placement and warming if POOR or ACQUIRING signal states persist for more than 5 continuous seconds.

6. End-to-End Application Lifecycle and Compliance

```
[ App Launch ]
      |
      v
[ Home Screen ] ---> (User Taps "Start Measurement")
      |
```

```
      +---> Create CSV Logger (Recursive creation of data/
directory)
      |
      +---> Request Camera Permission (Android / iOS Info.plist)
      |
      +---> Initialize Rear Camera -> Torch ON -> Wait 500 ms ->
Lock Exposure/Focus
      |
      v
[ Stream Active ] ---> Frame Processing Loop (YUV420 / BGRA8888
ROI Sampling)
      |
      +---> Stream Interruption Check (Gap > 100 ms? Reset
Resampler/DSP)
      |
      +---> State Evaluation: IDLE -> DETECTING -> ACTIVE ->
RELEASED
      |
      +---> Output Pipeline (If ACTIVE):
      |         |
      |         +--> Interpolate onto 30 Hz Grid
      |         +--> 0.75-3.5 Hz Bandpass Filtering
      |         +--> Peak Detection & Interval Validation
      |         +--> Update Displayed Waveform, Quality, & BPM
Meter
      |         +--> Append Raw Samples & Diagnostic Trace to CSV
      |
      v
[ Scan Termination Triggers ]
  1. Finger Removed: 3 bad frames -> 300 ms removal -> RELEASED
state.
  2. Inactivity Timeout: 20 seconds without ACTIVE status.
  3. Manual Tap: User selects "Finish" or "Close".
      |
      v
[ Scan Complete Screen ] (Camera/Torch Stopped -> Final Stats
Frozen)
      |
      v (User Taps "Done")
[ Return to Home Screen ]
```

Medical Disclaimer & Technical Scope

The application produces an **optical estimate** from low-resolution smartphone camera frames, not a medical-grade Electrocardiogram (ECG) or clinically validated diagnostic measurement. Signal artifacts—including periodic device motion, uneven finger pressure, ambient temperature changes, and torch thermal variations—can mimic real arterial pulses.

The output is intended solely for general tracking and educational demonstration, not for urgent medical decisions or clinical diagnosis.